Kildisiute 2021 neuroblastoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Malignant cells form a per-platform / per-patient archipelago. At the best-mixing integration (θ=10) the kNN graph is a single component — the scalar a naïve test reads as "integrated" — yet the embedding is organised BY SAMPLE: platforms NEVER share a neighbourhood (iLISI(technology)=1.00/2, pinned across the entire θ sweep 2–50), and at low θ the manifold is two disconnected blobs split cleanly by platform. The platform batch is perfectly confounded with patient (each patient on one platform only), so Harmony-on-sample has no anchor. Even within one platform the two patients barely mix (iLISI ≈1.8/2) — inter-patient / inter-platform (technical + CNV/clonal) heterogeneity, not a shared cell-state continuum.

Gate 2 — module geography → N/A

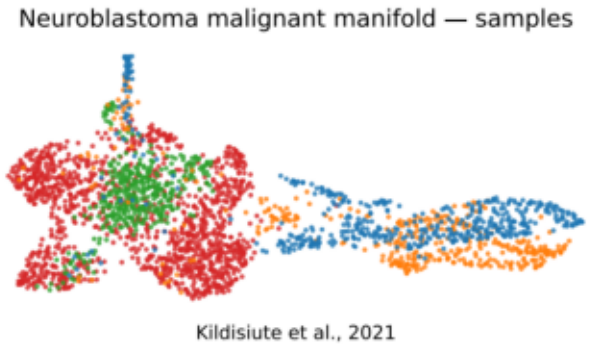
Not reached. Gate 2 tests a Gast/pSom hub and Crest geography on a fitted manifold; with Gate 1 NO-GO no manifold / trajectory is fit, so the module-geography gate does not apply.

Gate 3 — cross-repression → N/A

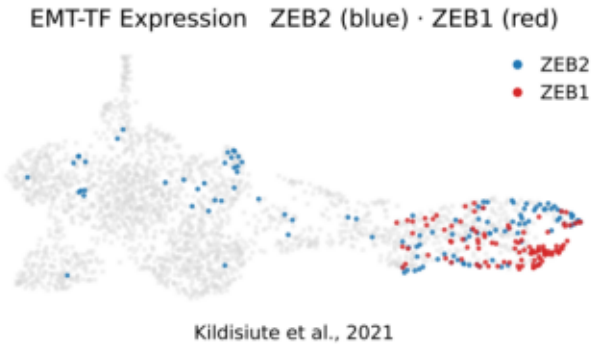
Not reached (branching-only, and downstream of Gate 2). No hub, no bifurcation, no hub r.

STEP FIGURES

Malignant manifold — samples



EMT-TF expression



Dataset.
Neuroblastoma (Kildisiute et al., 2021), Neuroendocrine — 10X (UMI) + CEL-seq2 (TPM) merged (inner join, 32,118 shared genes). ≥166-cell threshold keeps 4/14 samples (2,632 cells); a sparse 10-sample tail is dropped.

Verdict.
The embryonic-program pattern does NOT reproduce in Kildisiute 2021 neuroblastoma: the malignant compartment is a per-platform / per-patient archipelago with no sample-integrated, continuous cell-state manifold. Gate 1 NO-GO — analysis stops before the trajectory. ZEB2 / ZEB1 are the broadly-expressed EMT-TFs, reported unconditionally.

Parameter choices

Data merge	10X UMI + CEL-seq2 TPM, inner join (32,118 genes)	**Sample threshold**	≥166 cells (gap 412→67, 6.15×); keeps 4/14 (2,632 cells)
Gene / PCA	1,513 overdispersed; n_pcs=20 (PC1 ≈57% variance = CNV axis)	**Integration**	Harmony on sample, θ 2/5/10/20/50; θ=10 best; iLISI(tech)=1.00/2 throughout
Root / geometry	None — no trajectory fit (Gate 1 NO-GO)	**EMT-TFs shown**	ZEB2 (45.2%) · ZEB1 (44.4%); FOXC/PRRX added, VIM excluded